

Dear Referee,

Thank you for carefully reading the previous version of the paper and providing me with great comments and suggestions. I think that the paper has improved substantially thanks to your suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.
2. **Modeling.** A new “Key Assumptions” discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.
3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover’s rewards experiment shows some substitution at the margin. Appendix OF estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.
4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results (\$31 bn vs. \$27 bn).
5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 2

For clarity, I group my responses thematically rather than following the original numbering. I first address merchant-side concerns (points 1a and 2c on acceptance costs, and point 5 on merchant competition), then consumer-side modeling (points 1b and 6 on heterogeneity, multihoming, information sets, and debit-credit substitution including the Discover evidence), then market structure (point 1c on three- vs. four-party systems and points 2a–2b on counterfactual robustness), then data and measurement (points 2d, 3, and 4). Additional sub-points from your report are collected at the end.

Merchant acceptance costs: Unfortunately, not much has been done for the merchant side of the model. While equipment costs may be low, total acceptance costs matter. These can include installation fees, POS system integration, training, charge-back risks, and long-term contracts. Evidence suggests fixed costs are important, especially for small retailers. The paper should model or at least discuss these costs explicitly, as they can change equilibrium predictions dramatically.

Merchant acceptance costs and equilibrium fragility: With fixed costs, even minor changes could cause disintermediation. The paper should report a sequence of equilibria as fees are gradually reduced toward the cost of cash.

Reply: Fixed costs of card acceptance cannot be identified separately from heterogeneity in sales benefits (γ) without exogenous variation in consumer card adoption that shifts merchant acceptance. I acknowledge this limitation explicitly. The substitution-based specification in Online Appendix OF.3.1 pins down a fixed cost via an auxiliary cross-network acceptance ordering moment, but that is a calibration choice rather than a first-principles identification result, so I report it as a robustness check rather than a credible estimate.

Your concern about equilibrium fragility—that lower rewards could reduce consumer adoption enough to trigger cascading merchant disintermediation—directly shaped the choice of counterfactual. Rather than cap fees at the cost of cash, the main counterfactual now caps credit fees at 120 bp, roughly in line with Canadian merchant fees (Innovation 2023). This keeps the analysis in a range where acceptance cascades have not materialized in practice, so the predictions do not depend on extrapolating merchant behavior far from current equilibrium conditions.

In response to your suggestion to report a sequence of equilibria, Online Appendix OG.4 compares the moderate cap to more aggressive caps, including the cost-of-cash benchmark and the Ramsey planner’s solution. The relationship between cap level and welfare is concave: the 120 bps cap captures most of the planner’s welfare gains, and the cost-of-cash cap delivers only about 7% more. This concavity is reassuring from the perspective of your cascade concern, because it means the welfare conclusions do not depend on pushing fees into territory where disintermediation effects would dominate.

How does merchant competition in acceptance work if the benefit of accepting is the same for the first and last adopter?

Reply: Merchants differ in γ , which determines how much card acceptance increases sales to card-holders. A merchant accepts a card when the sales benefit from card-paying consumers exceeds the merchant fee, and this benefit is proportional to γ (Equation 12). The “first” adopter is the highest- γ merchant, for whom the sales gain far exceeds the fee; the “last” is the marginal merchant whose gain just covers it. There is an equilibrium in pure strategies where merchants above the threshold accept and those below stay cash-only, so competition works through the extensive margin of merchant acceptance rather than through identical merchants racing to adopt.

Consumer utility function and shopping choice: Why does mean unobserved utility of a card not vary with consumer type? Allowing heterogeneity only along monetary variables may bias results. Higher-income consumers likely have different bundles, and thus different unobserved utility. This could drive the finding that reward sensitivity rises with income. The model should allow $\Xi_{\omega i}$ to vary by income.

Why doesn’t χ vary with income or reward sensitivity? High-income and high-sensitivity consumers should gain more from multihoming.

Reply: I address these two comments together. The model allows unobserved utility to vary with consumer income through two channels. First, the random coefficient on card characteristics, $\beta_i \sim N(\beta_y \cdot \log \tilde{y}, \Sigma)$, means higher-income consumers have systematically different preferences for card characteristics. This allows the model to match average income differences between credit, debit, and cash users. What the model does not capture is whether high-income consumers have systematically different preferences

across specific networks like Visa, MC, and AmEx, because the most representative data with both income and payment choice (the Diary of Consumer Payment Choice) has limited AmEx coverage.

Second, in response to your concern about income-dependent multihoming, I have added income-dependent complementarity parameters $\chi_i^{lm} = \chi^{lm} + \chi_y^{lm} \log \tilde{y}$. This allows higher-income consumers to derive different value from holding multiple cards. This is the dimension of income heterogeneity that matters most for the welfare analysis. Multihoming patterns by income determine the competitive pressure merchants face and thus the equilibrium fees networks can charge. Allowing Ξ to also vary with income on top of β_y and χ_y would be difficult to identify, because the data are not rich enough to distinguish income differences in baseline network preferences from the variation these other channels already capture.

Information sets: The paper needs more discussion of what consumers know about merchant prices and acceptance decisions. If consumers observe both, why don't they sort perfectly across merchants? In the current model, sales benefits are fixed and independent of how many merchants accept. In reality, business stealing should reduce benefits as more merchants accept cards.

Reply: The model assumes consumers have complete information about merchant prices and acceptance decisions. This is now stated in Section IV.B.2: "Consumers observe merchants' acceptance decisions and prices before choosing where to shop."

Consumers do not sort perfectly across merchants because of CES preferences, not information frictions. All consumers carry cash. Payment acceptance only changes the intensive margin of where consumers shop. Consumers value variety across merchants, so even with full information they spread their spending rather than concentrating it at a single store.

On business stealing: the model does capture this. The sales benefit to an individual merchant from accepting cards depends on how many other merchants accept. When few merchants accept cards, a merchant that accepts captures a large share of card users' spending. As more merchants accept, card users spread their spending across more stores, reducing the benefit to any individual merchant. This shows up through the price index P^w , which depends on other merchants' acceptance decisions.

Debit-credit substitution: It seems unrealistic that consumers never substitute between debit and credit cards at the POS. Both are electronic, and often closer sub-

stitutes than either is with cash. Figure 4 suggests substitution from cash/debit to credit. The model should capture this, or at least acknowledge it.

Reply: This is a limitation. While incorporating credit-debit substitution would make the model more realistic along some dimensions, it also generates predictions at odds with the empirical evidence, so the main text keeps the assumption of point-of-sale segmentation.

The evidence on credit-debit substitution is mixed. On one hand, antitrust testimony in *Ohio v. AmEx* and consumer surveys describe credit and debit as distinct products at the point of sale (Online Appendix OF.1). On the other hand, Discover's rewards shock (Online Appendix OD.2) shifts consumer spending onto Discover from both cash and debit, consistent with some substitution at the point of sale.

Allowing substitution generates two equilibrium predictions (Online Appendix OF.2) at odds with industry commentary and data. The first is that credit acceptance would depend on credit-debit multi-homing rates, contradicted by *Ohio v. AmEx* testimony in which merchants discipline credit fees by threatening to route transactions to rival credit cards rather than debit. The second is that capping debit fees would lower credit fees, contradicted by Durbin's null on credit interchange (Figure 5).

Segmentation prioritizes the macro facts on merchant acceptance and pricing at the cost of imperfect fidelity to the micro evidence on consumer substitution. Online Appendix OF estimates two alternative specifications that navigate this trade-off differently. One allows point-of-sale substitution and incremental debit sales relative to cash (Online Appendix OF.3.1) and matches the micro evidence most closely. The other treats debit as generating no incremental sales relative to cash (Online Appendix OF.3.2) and is silent on Durbin pricing because debit fees are zero by construction. Total welfare effects are similar across all three for the credit fee cap (\$23–28B) and the merger (\$8–15B).

I do not understand footnote 12, which suggests that short-term transaction-specific incentives do not affect consumer usage decisions of credit versus debit cards, while rewards on all debit transactions do affect the adoption choice. Does Figure 5 apply only to consumers who can potentially substitute between Discover and debit (i.e., face the choice between them and cash) or is it an aggregate pattern across all consumers? Is it possible that the pattern on the figure is generated by two groups of consumers? Consumers in one group do not have any credit cards to substitute away from, while consumers in the second group have another credit card in addition to Discover making it easy to switch without using their debit cards. This question can probably be addressed by reporting the probability of observing

consumers with a debit card and Discover credit adoption combination only (i.e., without other cards from Visa or MC). If this group is relatively small or cannot be isolated for the purpose of the exercise, the evidence cannot be used in support of no substitution between debit and credit cards.

Reply: The role of the Discover event study in Online Appendix OD.2 is now narrower than in the previous draft. Its primary role is to show that consumers do not change where they shop in response to rewards, and that the null store-choice result is not driven by inattention. Table OD.3 confirms that consumers actively shift spending toward Discover during reward quarters (a 7.6 pp. increase) while their grocery spending share is flat.

To address your concern about consumer heterogeneity: the table below repeats the analysis for households whose primary or secondary card is Discover and who hold no other credit card among these top two cards, so they cannot easily substitute to another credit card. The control group consists of households who own only one credit card among their primary and secondary cards, and that card is not Discover. These households may own additional credit cards that they use less frequently. “Discover only” means Discover is the sole credit card among the primary and secondary cards recorded in the data.

The grocery spending share is again flat. The Discover spending share at grocery rises by only 2.5 pp ($p < 0.001$), compared to 7.6 pp in the full sample, consistent with debit being a much weaker substitute for credit than other credit cards are. My updated interpretation, described in Online Appendix OD, is that the Discover evidence suggests that consumers are willing, in a limited way, to substitute between credit and debit at the point of sale. Online Appendix OF discusses how to incorporate this evidence when thinking about credit-debit substitution at the point of sale.

Also, would it be possible to enforce debit card adoption if a consumer chooses to adopt a credit card such that the choice set contains options of being (1) cash-only user, (2) cash and debit user, (3) debit and one credit card user, and (4) debit and two credit cards user?

Reply: Your suggested choice set would eliminate consumers who use credit cards and cash but do not use debit cards. While most credit card holders technically own a debit card, the model captures usage patterns rather than ownership. The NielsenIQ Homescan data shows a meaningful share of consumers who use credit at virtually all stores but

Effect of Discover cashback rewards on payment composition: Discover-only credit card holders

	Grocery	Cash	Debit	Other Credit	Discover
Discover HH $\times$ Grocery $\times$ Reward Month		-0.005+ (0.003)	-0.017*** (0.004)	0.002 (0.002)	0.025*** (0.003)
Discover HH $\times$ Reward Month	-0.004+ (0.002)				
Observations	2741191	4586461	4586461	4586461	4586461
R ²	0.595	0.803	0.884	0.893	0.909
HH $\times$ Grocery FE		Yes	Yes	Yes	Yes
HH $\times$ Time FE		Yes	Yes	Yes	Yes
Grocery $\times$ Time FE		Yes	Yes	Yes	Yes
Household FE	Yes				
Time FE	Yes				
Clustered SE	HH	HH	HH	HH	HH

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Same specification as Table OD.3. Treatment: households whose only credit card among their primary and secondary cards is Discover (they may hold additional, less-used credit cards). Control: households with exactly one non-Discover credit card among their primary and secondary cards.

never use debit, even though they presumably have a bank account. Section IV.F (“Two-Card Restriction” paragraph) explains the data motivation: consumers put around 95% of their card spending on just two networks (Online Appendix Table OA.4), so the two-card model closely matches observed behavior.

Three- vs. four-party systems: Assuming away issuer/acquirer markets is not innocuous. AmEx may be more efficient as a three-party system. With regulation, Visa/MC could exit. The author should compute welfare if only AmEx survives after regulation.

Reply: Vertical relationships between networks, issuers, and acquirers are an important feature of payment markets. My model treats networks, issuers, and acquirers as a single entity that maximizes joint profits. As I now discuss in Section IV.F, this is motivated by the long-term contracting arrangements that tie these parties together in the U.S. market and by the DOJ’s 2024 antitrust suit against Visa, which alleges that the network deploys payments to issuers and potential rivals to coordinate the chain. Double marginalization and other vertical frictions between the three parties could still move fees and rewards quantitatively; a fully separated vertical model is an important direction for future work.

I do not compute the AmEx-only scenario directly because eliminating Visa and MC changes the number of logit draws consumers face, creating a variety-channel welfare effect that confounds the competitive mechanism. The merger counterfactual isolates competition cleanly: it holds the set of payment technologies fixed and changes only the competitive structure among networks. The Australian experience also speaks against exit: Australia imposed interchange fee caps on four-party networks in 2003, and neither Visa nor MC left the market, suggesting fee regulation reduces rents without pushing four-party networks out.

Counterfactual: What if credit cards are eliminated entirely? Is welfare higher than in the factual equilibrium? Edelman & Wright (2015) suggest consumers may be better off without intermediation. The author should explore this empirically, or at least trace equilibria as fees fall until cards vanish.

Reply: My model estimates imply that consumers are worse off by 13 billion dollars relative to an economy without card payments, consistent with Edelman and Wright (2015). But this number should not be taken literally: the random coefficients specification does not capture inframarginal consumers who place very high value on card features, so the welfare loss from full elimination depends heavily on modeling assumptions.

Rather than tracing equilibria all the way to zero, I compare the 120 bps cap to the cost-of-cash cap and the constrained social optimum. Online Appendix OG.4 shows that the cost-of-cash cap delivers only about 7% more welfare than the 120 bps cap, and the 120 bps cap captures roughly 77% of the Ramsey planner's welfare gains. The conclusions therefore do not require extrapolating to a world without cards.

Unobserved quality: Why should card quality remain fixed under regulation? Networks may cut services, degrade quality, or withdraw products. The author could compute how much deterioration would push usage near zero.

Reply: The model represents a short-run equilibrium in which unobserved card characteristics are held fixed. It is entirely fair that networks could adjust quality in response to regulation.

In practice, the Australian experience suggests that non-rewards annual fees did not rise after the fee caps, and the spread between credit card rates and unsecured term loans was unchanged (Appendix Figure OB.1). However, I cannot rule out that networks adjusted non-price characteristics that I do not measure, such as customer service,

fraud protection, or card benefits. I have added language in Section IV.F acknowledging this limitation and noting that the counterfactuals are best interpreted as short-run predictions.

Endogenizing product characteristics is a hard problem in its own right. Wollmann (2018) studies endogenous product entry in the truck market, and even in that simpler setting it requires substantial additional modeling and data. In the payment network context, it would require specifying how networks allocate resources across fraud protection, customer service, rewards structures, and other dimensions, which is beyond the scope of this paper.

New payment methods: If debit and credit are poor substitutes, a new method may not help. It would be good to discuss how such an entrant could gain traction.

Reply: You are correct. If debit and credit cards are poor substitutes, then a new payment method that resembles debit (such as many proposed public payment platforms) is unlikely to displace credit cards simply by offering lower fees. I clarify this in Section VI.E. This is consistent with the limited success of lower-fee payment alternatives in markets where credit cards are well-established.

Why is cash assumed cheapest? With contactless features, cards may be cheaper. The reliance on signature card data could bias unobserved quality estimates.

Reply: The model does not infer costs from the physical qualities of cards. Network costs are backed out from the networks' first-order conditions, and the cost of cash is taken from Felt et al. (2023), who measure resource costs of payment instruments in both Canada and the U.S.; I use their U.S. estimate.

The paper should clarify the years covered by each data source to ensure consistency.

Reply: The estimation appendix now states the years covered by each data source inline. Table OA.7 maps each data source to the parameters it identifies and lists additional sources used for validation.

In general, I would expect consumer price sensitivity to decline as income increases. Higher responsiveness of high-income consumers to rewards does not seem to require higher absolute price sensitivity if high income consumers spend substantially more on their cards. Due to the high total value of all transactions, the amount of pecuniary rewards is naturally higher for richer individuals. Hence, even with low price/reward sensitivity high income consumers should be gaining more from adopting and using cards than their low income counterparts.

Reply: Your point is correct that higher-income consumers earn more dollar rewards because they spend more, but α_i is sensitivity to the reward *rate*, not to dollar rewards. Pecuniary utility (Equation 16) depends on the reward rate f^j —a consumer who spends \$10,000 and one who spends \$1,000 face the same f^j . The income elasticity $\hat{\alpha}_y = 0.20$ (SE 0.06) is therefore identified from how wallet choices vary with income conditional on the reward rate, not from the mechanical correlation between income and total dollar rewards. This is consistent with evidence that higher-income consumers are more attentive to financial product terms (Hastings et al. 2017), even though they may be less sensitive to retail prices (Sangani 2024).

Figure 4, panel (b) “Effects on the number of trips” only illustrates the change in trips by credit card consumers. What happens to the number of trips by cash and debit card consumers? Did they decline by the same amount? This evidence is important in understanding where the increase in sales is coming from.

Reply: The triple-difference design conditions on consumers who use credit cards at other stores (treatment) versus those who do not (control), where the control group includes both cash-only and debit-only consumers. The coefficient therefore already measures the *differential* change in trips for credit card users relative to non-credit-card users at the same store, netting out any common trends, including changes in store-level traffic driven by concurrent pricing or demographic shifts.

For the structural model, the key moment is how credit card users respond to acceptance, which is what the triple-difference identifies regardless of whether cash and debit trips moved in parallel. Decomposing the raw trends by payment type would not be informative here, since those trends confound the store-level demand shocks that motivated the triple-difference design in the first place.

Differences between 3- and 4-party systems: the exercise of computing welfare if only AmEx survives after regulation could also illustrate the role of competition in rewards between card providers for the level of excessive card adoption.

Reply: Eliminating Visa and MC changes the number of logit shocks consumers draw over payment options, which affects welfare through a pure variety channel that is separate from the competitive mechanism you highlight. The merger counterfactual avoids this issue: it holds fixed the set of available payment technologies and changes only the competitive structure among networks, so the welfare comparison isolates the role of competition cleanly.

The author should be able to trace a sequence of equilibria for different levels of merchant fees starting from the factual level all the way until credit cards are no longer used in equilibrium. This potentially may require a reversal of the fees, with consumers instead of receiving rewards having to pay merchants for the opportunity to use credit cards.

Reply: Appendix OG.4 traces exactly this kind of sequence: I compare equilibria under the 120 bps cap, the tourist test cap (where merchant fees equal the cost of cash), and the Ramsey planner's solution. As you anticipated, at sufficiently low fee caps the networks do charge consumers negative rewards — that is, consumers pay for the use of cards. The welfare relationship is concave: most gains come from the initial fee reductions, and tighter caps yield diminishing returns.

Using data from only signature cards, which are less convenient and costlier to use than contactless ones, may lead to incorrect estimates of unobserved characteristics. If the data distinguishes between contactless and signature cards, it would be nice to see different estimates. Otherwise, the author may want to allow for some time variation in the unobserved product characteristics.

Reply: The data do not distinguish between contactless and signature transactions, only between debit and credit. The unobserved characteristics Ξ are estimated to rationalize observed market shares in a 2019 cross-section, so they absorb whatever mix of contactless and signature use prevailed at that time. Since the goal is to model the market structure as it stood in 2019, time variation in Ξ is not relevant to the exercise.

References

Edelman, Benjamin, and Julian Wright. 2015. "Price Coherence and Excessive Intermediation*." *The Quarterly Journal of Economics* 130, no. 3 (August): 1283–1328.

- Felt, Marie-Hélène, Fumiko Hayashi, Joanna Stavins, and Angelika Welte. 2023. "Regressive Effects of Payment Card Pricing and Merchant Cost Pass-through in the United States and Canada." *Journal of Banking & Finance* 154 (September): 106968.
- Hastings, Justine S., Ali Hortaçsu, and Chad Syverson. 2017. "Sales Force and Competition in Financial Product Markets: The Case of Mexico's Social Security Privatization." *Econometrica* 85 (6): 1723–1761.
- Innovation, Science and Economic Development Canada. 2023. "Government of Canada Announces Finalized Agreements with Visa and Mastercard to Lower Credit Card Transaction Fees for Small Businesses," December 5, 2023. News releases. Accessed January 26, 2026.
- Sangani, Kunal. 2024. "Markups Across the Income Distribution: Measurement and Implications." *Working Paper*.
- Wollmann, Thomas G. 2018. "Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles." *American Economic Review* 108, no. 6 (June 1, 2018): 1364–1406. Accessed February 17, 2026.